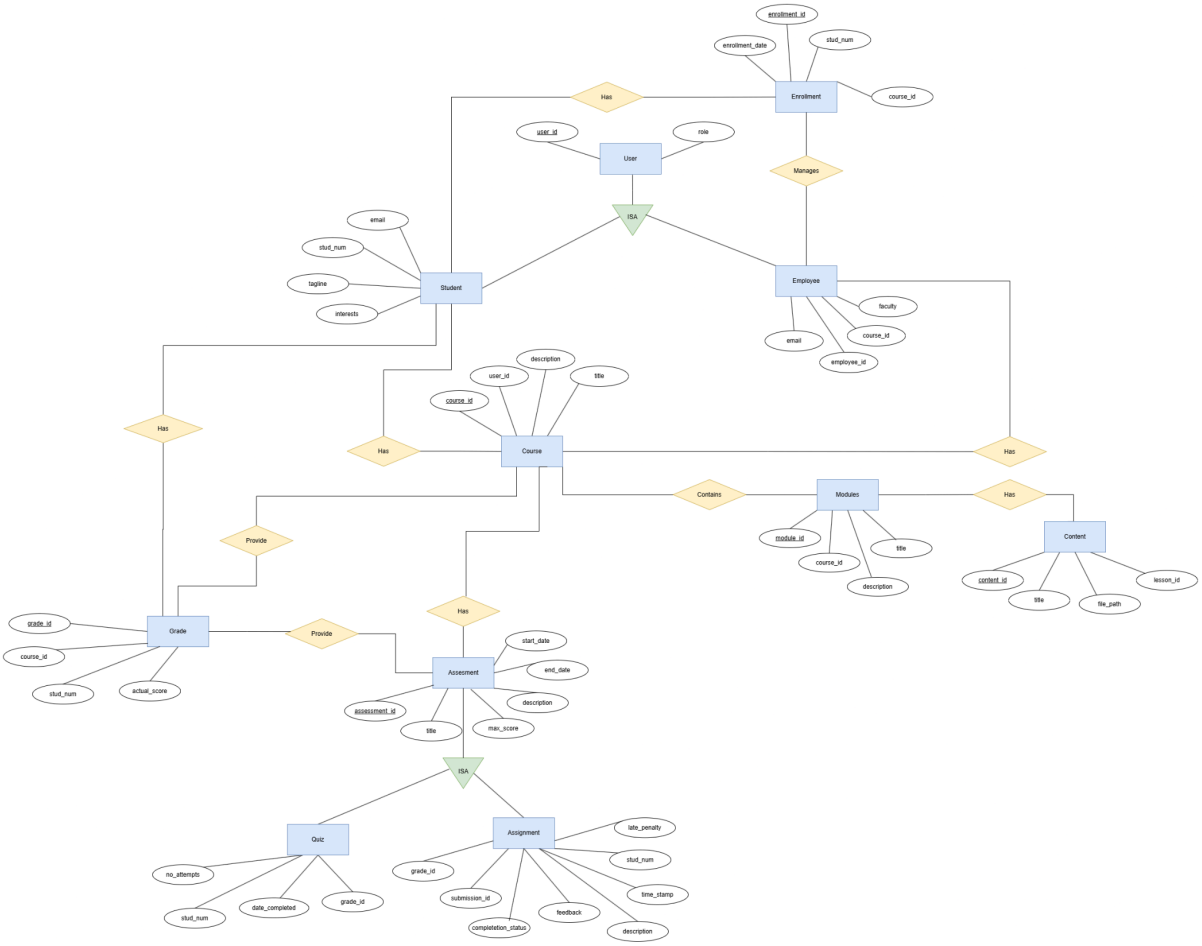
1. [Link to the diagram](#) (probably won't grant access)



2.

Student(<u>user_id</u>,stud_num, tagline, email, interests, role)

Enrollment(<u>enrollment_id</u>, enrollment_date, STUD_NUM, COURSE_ID)

Employee(<u>user_id</u>, email, employee_id, COURSE_ID, faculty, role)

Grade(<u>grade_id</u>, COURSE_ID, STUD_NUM, actual_score)

Course(<u>course_id</u>, USER_ID, description, title)

Module(<u>module_id</u>, COURSE_ID, description, title)

Content(<u>content_id</u>, title, file_path, LESSON_ID)

Quiz(<u>assessment_id</u>, title, max_score, description, start_date, end_date, no_attempts, STUD_NUM, date_completed, GRADE_ID)

Assignment(assessment_id, title, max_score, description, start_date, end_date, submission_id, completion_status, feedback, description, time_stamp, STUD_NUM, late_penalty, GRADE_ID)

Has(user_id, GRADE_ID)
Has(course_id, ASSESSMENT_ID)
Has(course_id, stud_num)
Has(stud_num, ENROLLMENT_ID)
Has(course_id, EMPLOYEE_ID)
Has(module_id, CONTENT_ID)
Manages(employee_id, ENROLLMENT_ID)
Provide(grade_id, COURSE_ID)
Provide(grade_id,ASSESSMENT_ID)
Contains(course_id, MODULE_ID)

3. Student: stud_num → tagline, email, interests.
   Each student number uniquely identifies a single student, meaning we can uniquely determine their tagline, email and interests from their student number.

4. In relation *Enrollment*: *enrollment_id* is the primary key and the relation contains *stud_num* and *email* which are related in the *Student* relation.
   Since enrollment_id → stud_num, and stud_num → email:
   enrollment_id → email.

5. The relationship: *has(course_id, assessment_id)* exists between the *Course* and *Assessment* relations. Here we have the composite key: *(course_id, assessment_id)*. From the *Assessment* entity: *assessment_id → title, max_score, description, start_date, end_date. assessment_id* is part of the aforementioned composite key, and since the right-hand side contains no primary keys, it is therefore a partial functional dependency.

6. Consider the following relations:
   - Student(user_id,stud_num, tagline, email, interests, role)
   - Course(course_id, USER_ID, description, title)
   - Enrollment(enrollment_id, enrollment_date, STUD_NUM, COURSE_ID)

   Here, the Enrollment relation provides a many-to-many relationship between Student and Course. That is, a Student can have many Courses and vice-versa. For Enrollment to be in 3NF, the following conditions must be met:
   - Relation must be in 2NF - Data is atomic and no partial dependencies
   - No transitive dependencies - each non-key attribute must be directly dependent on the primary key.

Observe that:

- Enrollment's primary key enrollment_id is atomic (not composite) and all of Enrollment's non-key attributes are fully functionally dependent on enrollment_id. This implies that Enrollment is in 2NF.
- There are no non-key attributes that are transitively dependent on enrollment_id. In other words, all non-key attributes are directly dependent on enrollment_id. For example, enrollment_data is not dependent on STUD_NUM nor COURSE_ID because knowing the value for either of these attributes cannot tell us what enrollment_date should be. A similar argument can be made for STUD_NUM and COURSE_ID. This, along with the previous point, implies that Enrollment is in 3NF.

7. *External file*

8.

a. Content.
The structure of the data in Content is more complex, is hierarchical and is more likely to evolve over time, compared to the data in Assessments. There's more relationships between the Content's data, and this makes it more suitable to be stored in a document-oriented database.

Example:


```
{
  "content": [
    {
      "Content_id": 360420,
      "Title": "Week 10 - State Machine Diagrams",
      "File_path": "Amathuba/d2l/le/lessons/360420/",
      "Lessons": [
        {
          "Lesson_id": 3075835,
          "Lesson_title": "State Machine Diagrams",
          "Start_date": "2025-04-29",
          "End_date": "2025-05-02"
        }
      ]
    }
  ]
}
```


b. Assessments.
The data in assessments is simple and doesn't have a complex structure. There is no deep nesting of data, and the meta-data contains mainly atomic values.

Keys & Values to be used:

The keys used should be the attributes such as assessment_id, title, max_score, description, start_date, and end_date. The values paired with the titles would be either Integers or Strings.
(“Assessment_id” : 104221
“Title” : ”Exercise 8 - State Machine Diagram”
“Max_score” : 15
“Start_date” : “2025-04-29”
“End_date” : “2025-05-06”)

9. We met to complete question 1 and 2, 8.

Group member	contribution
Rhadiya	3
Isa	4
Adam	7
Azaam	5
Kieran	6